MULTI-ORFT: Stable Online Reinforcement Fine-Tuning for Multi-Agent Diffusion Planning in Cooperative Driving

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***Abstract*—Closed-loop cooperative intersection planning requires jointly generating interaction-realistic multi-vehicle trajectories while optimizing safety and traffic efficiency under distribution shift. Diffusion planners pretrained by imitation can model multimodal interactions, but they are weakly aligned with closed-loop objectives. We propose MULTI-ORFT, which couples scene-conditioned diffusion pre-training with online reinforcement fine-tuning. Pre-training learns a multi-agent diffusion prior with dual-path scene conditioning: cross-attention injects structured context, and AdaLN-Zero modulates denoising features to improve map consistency in dense interactions. Post-training formulates a two-level MDP and uses VG-GRPO, a variance-gated group-relative policy optimization method, to turn grouped multimodal diffusion samples into stable online updates under multi-agent non-stationarity. On WOMD closed-loop benchmarks, MULTI-ORFT reduces collision and off-road rates and increases average speed over strong baselines and the pretrained policy; ablations validate conditioning, horizon design, and post-training data composition.**

***Index Terms*—Autonomous driving, diffusion policy, multi-agent planning, reinforcement learning, closed-loop simulation**

I. INTRODUCTION

Cooperative driving at intersections is a demanding multi-agent planning problem: vehicles must negotiate right-of-way, yield, merge, and proceed while continuously reacting to one another. The resulting joint behavior is stochastic and multi-modal, where multiple outcomes can be safe and reasonable. A practical planner must therefore generate *interaction-realistic* joint trajectories while optimizing safety and traffic efficiency in closed loop under distribution shift [1]–[3].

Diffusion planners provide a probabilistic way to model the *joint* distribution of multi-agent trajectories from demonstrations [4], [5]. The denoising process produces temporally consistent trajectories and supports multimodal sampling, which is attractive for interactive planning. However, imitation-only training is susceptible to objective misalignment and closed-loop distribution shift [6]: rare safety-critical events yield sparse supervision, and imperfect scene conditioning can produce trajectories that violate map geometry or interaction constraints [7]–[9]. In practice, this tension between *interaction modeling* and *constraint adherence* remains a key bottleneck for diffusion-based multi-agent planners.

RL post-training can better align pretrained planners with closed-loop objectives by coupling exploration with reward-driven optimization [10]–[14]. Unlike behavior cloning, RL can exploit closed-loop rollouts to expose rare failures and to shape trade-offs between safety and efficiency directly. Yet, most prior work is *offline* and does not interact with reactive environments, often yielding limited closed-loop gains [10], [14], [15]. Fully online RL is substantially harder in multi-agent planning due to non-stationarity from coupled interactions, compounding errors, and high-variance gradients when optimizing multi-step denoising chains.

To address these limitations, we propose MULTI-ORFT, which couples scene-conditioned diffusion pre-training with online RL post-training for cooperative planning. In pre-training, we build a multi-agent denoiser on Diffusion Transformers [16] and strengthen scene conditioning via a dual-path design: cross-attention injects contextual tokens, while AdaLN-Zero provides stable, step-aware feature modulation to improve map consistency under dense interactions. In post-training, we formulate a two-level MDP that couples the denoising chain with environment interaction, and propose VG-GRPO to convert grouped multimodal diffusion samples into robust online updates by gating groups and switching normalization based on within-group reward variance. The two phases are explicitly coupled: grouped multimodal diffusion samples provide within-group ranking signals, and VG-GRPO turns these signals into stable online improvements despite multi-agent non-stationarity.

Contributions: MULTI-ORFT couples a scene-conditioned multi-agent diffusion prior with online RL post-training for cooperative planning. It uses dual-path scene conditioning with cross-attention and AdaLN-Zero, and it formulates online post-training with a two-level MDP and VG-GRPO, a variance-gated group-relative optimizer. On the WOMD closed-loop benchmark (Table I), compared with SMART-large, MULTI-ORFT reduces CR from 2.22 to 1.91 and OR from 1.58 to 1.36, while increasing AS from 8.34 to 8.61. Ablations in Tables III–V quantify the impact of AdaLN-Zero, horizon ratio, and post-training data composition.

II. RELATED WORK

A. Multi-agent Behavior Modeling in Traffic Scenarios

Modeling the joint behaviors of multiple interacting agents is essential for advancing autonomous system, yet it remains challenging due to the inherent multimodality of vehicle trajectories [3], [17]. Typical distributional regression methods typically fit parametric continuous distributions such as Gaussian [18] , Laplace [19] to obtain a compact multimodal representation. Other studies incorporate goal anchors [20] or learnable intention queries [18], [21] into the decoding process

to generate multi-modal future motions. However, these designs often increase model complexity and memory usage, limiting scalability. Recently, generative models, notably autoregressive Transformer and diffusion models, have advanced multi-agent planning and simulation [11], [22], [23]. Autoregressive Transformer models cast multi-agent motion modeling as a next-token prediction task [24], [25], with methods such as SMART [22] learning categorical distributions over discrete motion tokens, and MotionLM [23] enabling weighted mode identification via pairwise sampling and a simple rollout aggregation. However, their discrete, sequential decoding can hinder temporal coherence and remain limited to partially joint dependencies. By contrast, diffusion models offer a compelling alternative, as they excel at modeling complex multimodal distributions while producing temporally consistent trajectories. They have been applied to decision-making, trajectory planning, and traffic simulation [26]–[28]. For example, CTG++ [29] employs a spatiotemporal Transformer that captures the evolving dynamics of multi-agent interactions, and VBD [7] combine diffusion trajectory model with behavior prediction to produce versatile traffic behaviors. Despite improved behavioral diversity, existing diffusion-based methods often struggle to balance agent interaction modeling with scene-conditioned consistency, leading to scene-inconsistent multi-agent trajectories [7]. Moreover, these methods are susceptible to closed-loop distribution shift and objective misalignment, limiting robustness and reliability in real-world deployment [9].

B. Reinforcement Post-training for Driving Planners

Reinforcement learning (RL) has been shown to further enhance the capabilities of pretrained models by combining sampling-based exploration with reward-driven policy optimization [9], [30], [31]. Recent studies on reinforcement learning for driving planners generally focus on fine-tuning two major classes of generative model. The first line of work, analogous to RL fine-tuning for large language models, uses autoregressive generation to model each motion token as a continuous distribution and perform policy improvement [12], [32], [33]. However, this approach inherently suffers from conflicts between sequence-level and token-level objectives, and temporal instability induced by sequential decoding. In contrast, diffusion models provide an inherently temporally consistent decision process and produce diverse actions through probabilistic denoising in continuous space, making them particularly well-suited to RL's exploration and exploitation paradigm. For example, TrajHF [13] proposes a human feedback-driven RL fine-tuning scheme that aligns generative trajectory models with diverse human driving preferences. ReCogDrive [14] fine-tunes the diffusion planner using RL with a non-reactive simulator to generate safer and more stable trajectories. Nevertheless, existing methods are predominantly offline and do not interact with reactive environments; consequently, limited rollouts and exploration often yield marginal performance gains [11], [13], [14]. Due to the uncertainty and compounding errors introduced by environment interaction, RL training in a fully closed-loop manner remains an open challenge [10], [14]. In this work, we develop a stable online RL post-training framework for diffusion-based planners to improve closed-loop planning performance and strengthen cooperative driving behavior.

III. Problem Statement

We study closed-loop cooperative driving at intersections, where a centralized planner acts as a traffic dispatcher: at each replanning cycle it observes the current multi-vehicle state and structured scene context, *jointly* proposes future plans for all participants, executes a short prefix, and replans from updated observations. Formally, a traffic scene is represented as $\mathcal{S} = (\mathbf{x}, \mathbf{u}, \mathbf{c})$, where $\mathbf{x} \in \mathbb{R}^{N_v \times T \times D_x}$ and $\mathbf{u} \in \mathbb{R}^{N_v \times T \times D_u}$ denote the state and control sequences of N_v vehicles, and $\mathbf{c} \in \mathbb{R}^{D_c}$ encodes scene context: (i) vehicle motion histories, (ii) HD map polylines, and (iii) traffic-light states. Our goal is to learn a scene-conditioned *joint* planning policy $\pi_\theta(\cdot \mid \mathbf{c})$ that generates realistic interaction behaviors while improving closed-loop *safety* and *efficiency*.

This problem is challenging due to multimodal interaction under a high-dimensional joint distribution, hard scene constraints, and distribution shift with sparse safety-critical supervision. We address it with a two-phase pipeline: the pre-training phase (Sec. IV) learns a scene-conditioned diffusion prior from expert demonstrations to model multimodal cooperative behaviors; the online post-training phase (Sec. V) aligns the diffusion policy with closed-loop objectives by maximizing the expected cumulative reward $\bar{\mathcal{J}}(\pi_\theta)$.

IV. Multi-Agent Diffusion Prior for Cooperative Planning

We model joint multi-vehicle control chunks with a diffusion policy conditioned on structured scene context. Our key addition is dual-path conditioning that combines cross-attention with AdaLN-Zero to improve scene consistency.

A. Model Structure of the Multi-Agent Diffusion Planner

As shown in Fig. 1, the planner consists of a symmetric scene encoder $\mathcal{E}$ and a denoising decoder $\mathcal{D}_\theta$. The encoder maps raw context $\mathbf{c}$ to latent tokens $\tilde{\mathbf{c}} = \mathcal{E}(\mathbf{c})$ using query-centric attention in local frames [18]. Conditioned on $\tilde{\mathbf{c}}$ and diffusion step k, the decoder denoises joint controls $\mathbf{u}_k \to \mathbf{u}_0$ with inter-agent self-attention and dual-path scene conditioning with cross-attention and AdaLN-Zero. This design preserves multimodal interaction capacity while improving geometric constraint adherence.

B. Symmetric Scene Context Modeling Encoder

We tokenize $\mathbf{c}$ into agent-history tokens, map-polyline tokens, and traffic-light tokens, and encode them with a query-centric Transformer in local coordinates [18]. Vehicle-centered and polyline-centered frames normalize relative geometry and improve invariance to global pose. This design symmetrically aggregates relations among heterogeneous scene elements, yielding a unified context embedding $\tilde{\mathbf{c}} \in \mathbb{R}^{(N_v+N_m+N_t)\times D}$ for downstream denoising.

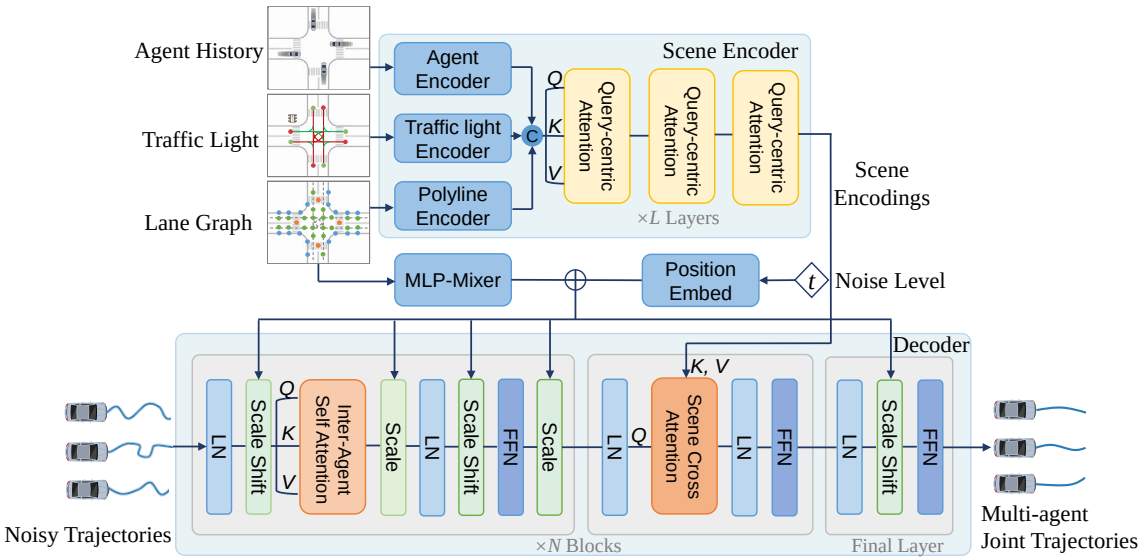


Fig. 1. Architecture of the multi-agent diffusion planner. A symmetric scene encoder captures local relational context with query-centric self-attention, and a denoising decoder generates joint multi-agent plans via inter-agent attention and scene-conditioned modulation with AdaLN-Zero.

C. Multi-Agent Denoising Decoder

We diffuse joint control chunks in the action space of all vehicles and learn a denoiser $\mathcal{D}_\theta$ conditioned on $\tilde{\mathbf{c}}$ and diffusion step k. We embed noisy trajectories by rolling out controls through dynamics $f(\cdot)$ to obtain states $\mathbf{x}$ and encoding them with an MLP; this keeps kinematic structure explicit during denoising. The denoiser then applies Transformer blocks with inter-agent self-attention.

Scene conditioning via cross-attention and AdaLN-Zero. Cross-attention alone can underutilize scene context under strong interaction coupling, leading to constraint violations [7]. We therefore couple (i) *cross-attention*, which injects $\tilde{\mathbf{c}}$ as keys/values to update trajectory tokens, with (ii) *AdaLN-Zero*, which provides stable layer-wise modulation conditioned on the diffusion step and dense road features. Specifically, since road-graph embedding $\tilde{\mathbf{c}}_{mp}$ can be sparse, we first densify it using a lightweight MLP-Mixer block that iteratively mixes information across the token and feature dimensions.

$$\begin{aligned} \tilde{\mathbf{c}}_{mp} &\leftarrow \tilde{\mathbf{c}}_{mp} + \mathrm{MLP}_{\text{tok}}\big(\tilde{\mathbf{c}}_{mp}^{\top}\big)^{\top}, \\ \tilde{\mathbf{c}}_{mp} &\leftarrow \tilde{\mathbf{c}}_{mp} + \mathrm{MLP}_{\text{ch}}(\tilde{\mathbf{c}}_{mp}). \end{aligned} \tag{1}$$

AdaLN then modulates each trajectory-token feature $\mathbf{h}$ by

$$\mathrm{AdaLN}(\mathbf{h}) = \big(1 + \lambda(k, \tilde{\mathbf{c}}_{mp})\big) \odot \mathbf{h} + \delta(k, \tilde{\mathbf{c}}_{mp}), \tag{2}$$

where $\lambda(\cdot)$ and $\delta(\cdot)$ are the scale and shift factors regressed from the sum of the diffusion-step embedding and the dense road features. We further regress a gating factor $\alpha(k, \tilde{\mathbf{c}}_{mp})$ applied to the residual branch. We initialize all α to zero so that the full decoder starts as an identity function and gradually learns to introduce conditioning, which strengthens conditional modulation and improves training stability.

Finally, the noise embedding is propagated to the final layer, which outputs the denoised action sequence $\mathbf{u}_0$. We train the decoder $\mathcal{D}_\theta(\mathbf{u}_k, k, \tilde{\mathbf{c}})$ by minimizing the following loss:

$$\mathcal{L}_{\mathcal{D}_\theta} = \mathbb{E}_{\mathbf{u}_0,\, k,\, \mathbf{u}_k \sim p_k(\cdot|\mathbf{u}_0)} \left[\mathcal{SL}_1(\mathbf{x}(\mathcal{D}_\theta(\mathbf{u}_k, k, \tilde{\mathbf{c}})) - \mathbf{x}_{gt})\right], \tag{3}$$

where $\mathcal{SL}_1(\cdot)$ is the Smooth-ℓ_1 loss between the expert trajectory $\mathbf{x}_{gt}$ and the trajectory $\mathbf{x}(\mathbf{u})$ obtained by rolling out a control chunk $\mathbf{u}$ through the dynamics $f(\cdot)$.

V. Planning-oriented Online Reinforcement Learning

A multi-agent diffusion planner trained solely via imitation learning can capture general driving capabilities, but it often fails under closed-loop execution due to distribution shift and the scarcity of safety-critical interaction data. Moreover, it fails to explicitly promote safety- and efficiency-oriented cooperative behaviors, relying solely on the BC loss.

To overcome these limitations, we propose a stable *online* RL post-training framework (Fig. 3) that enhances both robustness and cooperation in closed-loop planning. To support online optimization, we first formulate a two-level MDP. We then jointly design dense, well-shaped rewards and a variance-gated group-relative optimization (VG-GRPO) strategy to address instability during closed-loop training. Finally, we present the overall RL post-training pipeline.

A. Two-Level MDP

We now tailor the MDP to diffusion-policy optimization. Unlike standard policies that sample a_t in one shot, a diffusion (chunked) policy produces the executed action $\mathbf{u}_t^0$ through a K-*step stochastic denoising chain*. Treating the entire denoising chain as a black-box sampler in a single-level MDP leads

to poor credit assignment across denoising steps and high-variance gradients, which is exacerbated in *multi-vehicle online* rollouts due to severe non-stationarity from coupled agent interactions. We therefore model denoising as an inner MDP $\mathcal{M}_{\text{DP}}$ with *analytically tractable Gaussian likelihoods* at each step, coupled with an outer environment-interaction MDP $\mathcal{M}_{\text{ENV}}$ that provides closed-loop rewards and constraints. Following [34], their composition yields the two-level MDP in Fig. 2. We use subscript t for the outer environment step and superscript k for the inner denoising step.

- **State.** $\bar{\mathbf{s}}_t^k = (\mathbf{c}_t, \mathbf{u}_t^k)$, where $\mathbf{c}_t$ is the scene observation and $\mathbf{u}_t^k$ is the denoising state at step k. We denote the encoded context by $\tilde{\mathbf{c}}_t = \mathcal{E}(\mathbf{c}_t)$.
- **Action.** For $k > 0$, $\bar{\mathbf{a}}_t^k = \mathbf{u}_t^{k-1}$ sampled from the step-wise reverse transition $p_\theta(\mathbf{u}_t^{k-1} \mid \mathbf{u}_t^k, \tilde{\mathbf{c}}_t)$, which together with $\mathcal{N}(\mathbf{u}_t^K; 0, \mathbf{I})$ induces the diffusion-policy marginal $\pi_\theta(\mathbf{u}_t^0 \mid \tilde{\mathbf{c}}_t)$ executed in the outer MDP.
- **Initial distribution.** $\bar{\mathbf{P}}_0$, where $\mathbf{c}_0 \sim P_0$ and $\mathbf{u}_0^K \sim \mathcal{N}(0, \mathbf{I})$.
- **Transition.** The kernel $\bar{\mathbf{P}}(\bar{\mathbf{s}}_{t'}^{k'} \mid \bar{\mathbf{s}}_t^k, \bar{\mathbf{a}}_t^k)$ is

$$\bar{\mathbf{s}}_{t'}^{k'} = \begin{cases} (\mathbf{c}_t, \bar{\mathbf{a}}_t^k) & (t', k') = (t, k-1),\ k > 0, \\ (\mathbf{c}_{t+1}, \mathbf{u}_{t+1}^K) & (t', k') = (t+1, K),\ k = 0, \end{cases} \tag{4}$$

  where for $k = 0$ we execute $\mathbf{u}_t^0$ in the simulator to obtain $\mathbf{c}_{t+1}$ and re-initialize $\mathbf{u}_{t+1}^K \sim \mathcal{N}(0, \mathbf{I})$.
- **Reward.** Rewards are assigned only when denoising reaches $k = 0$:

$$\bar{R}(\bar{\mathbf{s}}_t^k, \bar{\mathbf{a}}_t^k) = \begin{cases} 0 & k > 0, \\ R_{\text{ENV}}(\mathbf{c}_t, \mathbf{u}_t^0) & k = 0, \end{cases} \tag{5}$$

  where R_{ENV} is specified in Sec. V-B.
- **Objective.** The outer-step objective is

$$\bar{\mathcal{J}}(\pi_\theta) = \mathbb{E}_{\pi_\theta, \bar{\mathbf{P}}, \bar{\mathbf{P}}_0} \left[\sum_{t=0}^{\infty} \gamma^t R_{\text{ENV}}(\mathbf{c}_t, \mathbf{u}_t^0) \right]. \tag{6}$$

- **Step-wise likelihood.** Each inner-step conditional is Gaussian:

$$\begin{aligned} p_\theta(\bar{\mathbf{a}}_t^k \mid \bar{\mathbf{s}}_t^k) &= p_\theta(\mathbf{u}_t^{k-1} \mid \mathbf{u}_t^k, \tilde{\mathbf{c}}_t) \\ &= \mathcal{N}\big(\mathbf{u}_t^{k-1};\ \mu_\theta(\mathbf{u}_t^k, \tilde{\mathbf{c}}_t, k),\ \sigma_k^2 \mathbf{I}\big). \end{aligned} \tag{7}$$

  where $\mu_\theta(\cdot)$ follows the standard DDPM closed form [35] using $\hat{\epsilon} = \mathcal{D}_\theta(\mathbf{u}_t^k, k, \tilde{\mathbf{c}}_t)$, and σ_k^2 follows the fixed noise schedule. Hence, the step-wise log-likelihood is analytically tractable.

B. Reward Modeling

Rewards determine both optimization direction and driving preference. We use a simple rule-based reward that generalizes across large-scale mixed scenarios and follows a safety-first principle with an explicit efficiency term. The outer-step environment reward is defined as

$$R_{\text{ENV}}(\mathbf{c}_t, \mathbf{u}_t^0) = w_c\, R_{\text{collision}}(t) + w_o\, R_{\text{offroad}}(t) + w_e\, R_{\text{eff}}(t). \tag{8}$$

We set $w_c, w_o \gg w_e \geq 0$ to prioritize safety.

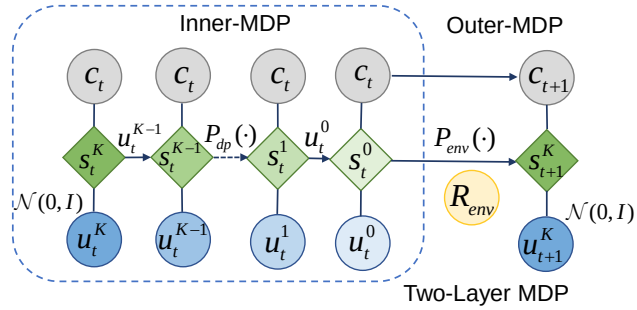


Fig. 2. Two-level MDP for online RL. An inner denoising MDP is coupled with an outer environment-interaction MDP, and step-wise Gaussian likelihoods make denoising-step policy gradients tractable.

Collision. We define collision by SAT-based oriented-box overlap:

$$R_{\text{collision}}(t) = -\mathbb{1}[\text{coll_overlaps}(t)]. \tag{9}$$

Off-road. We define off-road events from signed distances of the four vehicle-box corners to lane boundaries:

$$R_{\text{offroad}}(t) = -\mathbb{1}[\text{off_road}(t)]. \tag{10}$$

Efficiency. We measure normalized progress along a reference centerline:

$$R_{\text{eff}}(t) = \max\left(\frac{s_t - s_{t-1}}{s_{\max}}, 0 \right), \tag{11}$$

where s_t is the arc-length projection of the vehicle position onto its centerline and $s_{\max}$ normalizes progress to $[0, 1]$. In multi-vehicle rollouts, we aggregate R_{ENV} by averaging over controlled vehicles at step t. This formulation provides stable learning signals and explicitly enforces the intended trade-off between safety and efficiency. To avoid dynamic-quality collapse when emphasizing safety, we use three built-in safeguards rather than ad-hoc post-filters: dynamics-aware action-space denoising with rollout through $f(\cdot)$ (Sec. IV-C), KL anchoring to the pre-training prior in VG-GRPO, and short-horizon execute-then-replan in closed loop.

C. Variance-Gated Group Relative Policy Optimization

With the two-level MDP (Sec. V-A) and reward (Sec. V-B) in place, we optimize θ to maximize the outer-step objective $\bar{\mathcal{J}}(\pi_\theta)$, where π_θ is the marginal diffusion policy induced by the reverse denoising chain [35]. We propose Variance-Gated Group-Relative Policy Optimization (VG-GRPO) for multi-vehicle online diffusion-policy refinement. Our design is built on the critic-free GRPO paradigm [36], [37] and extends it with variance-gated advantages and denoising-aware optimization to stabilize training under multi-agent non-stationarity. This is a functional coupling rather than a loose combination: diffusion provides grouped multimodal candidates each step, and VG-GRPO directly consumes their relative ranking signal for online updates.

Group sampling. At each outer environment step t, we fix the encoded context $\tilde{\mathbf{c}}_t$ and sample a group of G candidate action chunks by running the inner denoising chain under the behavior policy $p_{\theta_{\text{old}}}$. Let $i \in \{1, \ldots, G\}$ index samples in the group. Each sample yields an executed clean chunk $\mathbf{u}^0_{t,i}$ and a scalar group reward

$$r_{t,i} := R_{\text{ENV}}(\mathbf{c}_t, \mathbf{u}^0_{t,i}), \tag{12}$$

i.e., we set the rollout horizon to $H = 1$ in this group-relative update. This produces a within-group reward set $\{r_{t,i}\}_{i=1}^G$.

VG-GRPO objective. For inner denoising step $k \in \{1, \ldots, K\}$, we define the likelihood ratio

$$\rho^k_{t,i}(\theta) := \frac{p_\theta\big(\mathbf{u}^{k-1}_{t,i} \mid \mathbf{u}^k_{t,i}, \tilde{\mathbf{c}}_t\big)}{p_{\theta_{\text{old}}}\big(\mathbf{u}^{k-1}_{t,i} \mid \mathbf{u}^k_{t,i}, \tilde{\mathbf{c}}_t\big)}, \tag{13}$$

where $p_\theta(\mathbf{u}^{k-1} \mid \mathbf{u}^k, \tilde{\mathbf{c}}_t)$ is the Gaussian reverse transition in Sec. V-A. We use a PPO-style clipped surrogate with a denoising-step discount $\gamma_{\text{denoise}} \in (0, 1]$:

$$\begin{aligned}\mathcal{L}^k_{t,i}(\theta) = -\beta\, \widehat{D}_{KL}(p_\theta \,\|\, p_{\text{ref}}) + \min\Big(&\rho^k_{t,i}(\theta)\, \gamma^k_{\text{denoise}}\, A_{t,i}, \\ &\text{clip}(\rho^k_{t,i}(\theta), 1-\varepsilon, 1+\varepsilon)\, \gamma^k_{\text{denoise}}\, A_{t,i}\Big),\end{aligned} \tag{14}$$

The overall surrogate objective is

$$\mathcal{J}_{\text{VG}}(\theta) = \mathbb{E}_{(\mathbf{c}_t, \{\mathbf{u}^{0:K}_{t,i}\}_{i=1}^G) \sim \mathcal{D}_{\theta_{\text{old}}}} \left[\frac{1}{G} \sum_{i=1}^{G} \frac{1}{K} \sum_{k=1}^{K} \mathcal{L}^k_{t,i}(\theta) \right], \tag{15}$$

where ε is the clipping range. Unless stated otherwise, we set $\gamma_{\text{denoise}} = 1$.

We set the reference policy to the pre-training output, i.e., $p_{\text{ref}} = p_{\theta_0}$, to preserve pre-trained driving behaviors.

KL regularization. We use the non-negative unbiased estimator in [38] at each denoising step:

$$\begin{aligned}\widehat{D}_{KL}(p_\theta \,\|\, p_{\text{ref}}) = &\frac{p_{\text{ref}}\big(\mathbf{u}^{k-1}_{t,i} \mid \mathbf{u}^k_{t,i}, \tilde{\mathbf{c}}_t\big)}{p_\theta\big(\mathbf{u}^{k-1}_{t,i} \mid \mathbf{u}^k_{t,i}, \tilde{\mathbf{c}}_t\big)} \\ &- \log \frac{p_{\text{ref}}\big(\mathbf{u}^{k-1}_{t,i} \mid \mathbf{u}^k_{t,i}, \tilde{\mathbf{c}}_t\big)}{p_\theta\big(\mathbf{u}^{k-1}_{t,i} \mid \mathbf{u}^k_{t,i}, \tilde{\mathbf{c}}_t\big)} - 1,\end{aligned} \tag{16}$$

where $\beta \geq 0$ controls the regularization strength.

Variance-gated advantages. Standard within-group normalization can amplify noise when the reward variance is small. We therefore gate advantage computation by the within-group standard deviation, inspired by [39]. Let the within-group reward statistics be

$$\mu_r = \text{mean}\big(\{r_{t,i}\}_{i=1}^G\big), \qquad \sigma_r = \text{std}\big(\{r_{t,i}\}_{i=1}^G\big). \tag{17}$$

We define the group-relative advantage for sample i as

$$A_{t,i} = \begin{cases} \text{drop this group}, & \sigma_r \leq \text{std}_{\text{low}}, \\ r_{t,i} - \mu_r, & \text{std}_{\text{low}} < \sigma_r \leq \text{std}_{\text{high}}, \\ \dfrac{r_{t,i} - \mu_r}{\sigma_r}, & \sigma_r > \text{std}_{\text{high}}. \end{cases} \tag{18}$$

In our implementation, $A_{t,i}$ is computed from $\{r_{t,i}\}_{i=1}^G$ and shared across all denoising steps k within the same chain. If a group is dropped in Eq. (18), we skip its update and compute Eq. (15) over the remaining groups. Extremely low-variance groups provide little ranking signal and are discarded; moderate-variance groups use unnormalized differences to avoid noise amplification; high-variance groups use standard normalization for scale control. This variance gate improves gradient quality and stabilizes multi-vehicle online diffusion-policy refinement while retaining informative samples.

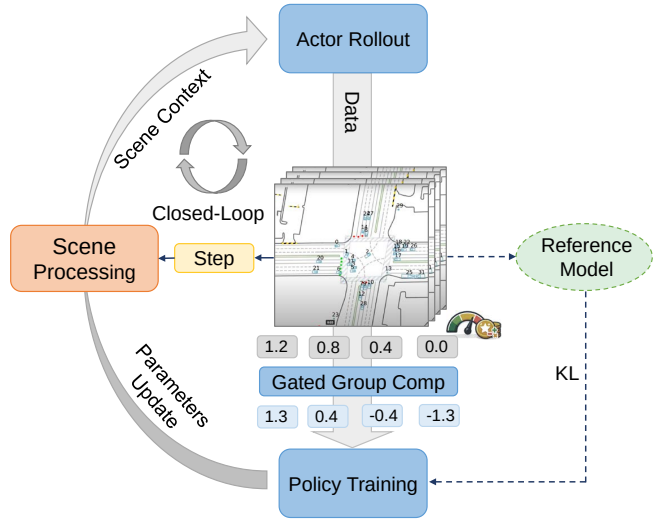


Fig. 3. Online RL post-training framework with three components: policy rollout, rule-based reward evaluation with group-relative advantages, and policy optimization with KL regularization.

D. Reinforcement Learning Post-training Framework

We implement the online RL framework in Fig. 3 with three stages: rollout, reward evaluation, and policy update. At each outer step t, MADP samples a group of G candidate chunks from the latest context $\tilde{\mathbf{c}}_t$, scores each chunk by rule-based safety and efficiency rewards, and updates the denoiser with **VG-GRPO** (Sec. V-C) plus KL regularization to the pre-training prior. We use a rolling horizon (plan T_p, execute first T_a), execute the best-performing sample for environment interaction, and retain *all* group samples for relative-advantage computation.

We store grouped tuples in the rollout buffer $\mathcal{D}_{\text{itr}}$:

$$\Big(\{\bar{\mathbf{s}}^k_{t,i}\}_{i=1}^G,\ \{\bar{\mathbf{a}}^k_{t,i}\}_{i=1}^G,\ \{r_{t,i}\}_{i=1}^G,\ k\Big), \tag{19}$$

where $\bar{\mathbf{s}}^k_{t,i} = (\mathbf{c}_t, \mathbf{u}^k_{t,i})$ and $\bar{\mathbf{a}}^k_{t,i} = \mathbf{u}^{k-1}_{t,i}$ follow the two-level MDP in Sec. V-A. During updates, we sample mini-batches from $\mathcal{D}_{\text{itr}}$, compute variance-gated group-relative advantages (Eq. (18)), and optimize the denoiser parameters by maximizing the VG-GRPO objective (Eq. (15)) with KL regularization to the pre-training reference policy.

VI. Simulation Results and Discussion

This section reports closed-loop evaluation of the proposed planner. We first describe datasets, metrics, and implementation details, then present benchmark comparisons and ablation

studies. The evaluation verifies three claims: closed-loop safety and efficiency gains; robustness of the online post-training design choices; and the contribution of the AdaLN-Zero conditional module.

A. Setup

Dataset. For pre-training, we use the Waymo Open Motion Dataset (WOMD) [40], containing 486,995 training scenarios and 44,097 validation scenarios; each scenario covers 9 s of real traffic with full participant trajectories and map topology. For closed-loop evaluation, we use 41,590 highly interactive test scenarios. For post-training, we uniformly sample 20,756 scenarios from the WOMD high-interaction validation split and ensure this set is disjoint from the closed-loop test set. We score these scenarios with the pre-training planner and construct three subsets: a low-score set of 2,504 failure-heavy scenarios with collisions and off-road events, a high-score set of 5,857 easy scenarios, and a full mixed set.

Implementation Details. Scene inputs include up to $N_v = 32$ agents, $N_m = 256$ polylines with $M_w = 30$ waypoints each, and $N_t = 16$ traffic lights. The planner predicts $T_p = 80$ control steps at 0.1 s resolution (using up to 1 s history during training); for closed-loop inference we condition on the current state only to reduce temporal leakage. In closed-loop testing and RL rollouts, we execute $T_a = 10$ steps and replan with a rolling horizon.

The scene encoder uses $L = 6$ query-centric Transformer layers with hidden size $D = 256$. The denoising decoder alternates two block types for three rounds (six Transformer layers total). MLP-Mixer token/channel dimensions are 64/128. Pre-training uses the logarithmic noise schedule in [7] with $\bar{\alpha}_{\min} = 10^{-9}$, scaling 0.0031, and $K = 20$. We pre-train with AdamW (weight decay 0.01), learning rate 2×10^{-4} (3000-step warmup; $\times0.02$ decay every 3000 steps), gradient clipping 1.0, BF16 on 4×RTX 4090 for 30 epochs (batch size 32).

Post-training uses 3 parallel rollout environments and group size $G = 10$. Each rollout batch is optimized for one epoch with mini-batch size 16 for 10M fine-tuning steps. Following [9], we use 4 s reward horizon, reward weights (collision/off-road/progress) $8/1/4$, KL weight $\beta = 0.2$, and learning rate 10^{-5}. For exploration and stability we use DAPO-style asymmetric clipping $[-0.15, 0.2]$ [39], $\sigma^{\text{sam}}_{\min} = 0.2$, $\text{std}_{\text{low}} = 0.1$, $\text{std}_{\text{high}} = 0.2$, $\sigma^{\text{prob}}_{\min} = 0.1$, and $\gamma_{\text{denoise}} = 0.9$. Post-training runs in BF16 on one NVIDIA 5090 GPU.

Metrics. We report collision rate CR, off-road rate OR, average speed AS, average displacement error ADE, and kinematic infeasibility Kin. We treat CR, OR, and AS as *primary* closed-loop objectives reflecting safety and traffic efficiency, and use ADE and Kin as *secondary* diagnostics for imitation fidelity and physical feasibility. CR uses SAT-based oriented-box overlap; OR checks drivable-area boundary crossing; AS is mean per-step displacement; ADE is the ℓ_2 position error to ground truth; and Kin counts violations of acceleration and curvature bounds with limits $6\,\text{m/s}^2$ and $0.3\,\text{m}^{-1}$. For fairness, all methods are evaluated with the same simulator configuration, including scenario split, agent count, horizon, and replanning frequency; values with $\pm$ report mean and standard deviation over repeated closed-loop evaluations in the main comparison tables, while ablation tables report point estimates under the same protocol for compactness.

TABLE I
CLOSED-LOOP BENCHMARK ON THE WOMD INTERACTIVE TEST SPLIT (PRIMARY: CR, OR, AS; SECONDARY: ADE, KIN)

Method	Primary closed-loop objectives			Secondary diagnostics	
	CR (%)↓	OR (%)↓	AS (m/s)↑	ADE (m)↓	Kin (%)↓
TrafficBotsV1.5 [41]	2.74±0.21	1.79±0.14	8.03±0.48	1.68±0.09	0.26±0.02
SMART-large [22]	2.22±0.09	1.58±0.10	8.34±0.30	1.30±0.01	**0.21±0.01**
VBD [7]	2.46±0.14	1.92±0.18	8.08±0.52	1.41±0.02	0.24±0.01
SMART-tiny-CLSFT [42]	2.10±0.10	1.53±0.12	8.47±0.44	**1.23±0.03**	0.25±0.02
MULTI-ORFT	**1.91±0.12**	**1.36±0.08**	**8.61±0.46**	1.36±0.04	0.32±0.03

B. Evaluation

This subsection evaluates interactive trajectory generation using both benchmark tables and representative qualitative cases.

WOMD Results. We compare four open-source baselines: VBD [7], SMART-large [22], SMART-tiny-CLSFT [42], and TrafficBotsV1.5 [41]. They cover autoregressive, behavior-cloning, and diffusion paradigms. Each scenario runs for 8 s in closed loop with 32 simulated agents at 1 Hz replanning. Table I shows that MULTI-ORFT leads on all three primary closed-loop objectives. Compared with SMART-large, it reduces CR from 2.22 to 1.91 (-14.0%) and OR from 1.58 to 1.36 (-13.9%), while increasing AS from 8.34 to 8.61 (+3.2%). Relative to the pretrained policy in Table II, online post-training yields 5.4% lower CR, 19.0% lower OR, and 3.0% higher AS. This is consistent with the mechanism: VG-GRPO directly optimizes safety and efficiency rewards in closed loop, enabling better interaction outcomes under distribution shift.

Post-training Methods. We further compare four post-training strategies, SFT, DPO, offline RL, and online RL, as shown in Table II. All methods improve over the pre-training-only policy in part, but with different objective alignment. SFT mainly improves imitation-oriented diagnostics (ADE). DPO and offline RL improve selected indicators but can degrade the balance between safety and efficiency. MULTI-ORFT with online RL provides the most consistent co-improvement on the primary closed-loop objectives CR, OR, and AS, reflecting the benefit of coupling diffusion group sampling with VG-GRPO.

Online RL shows a mild ADE increase, indicating a distributional trade-off: stronger closed-loop exploration can reduce one-step imitation accuracy while improving interaction robustness. Kin remains low across methods, all below 0.4%; compared with the pretrained policy, it increases by at most 0.07 percentage points from 0.25 to 0.32, indicating no dynamic-quality collapse under safety-oriented optimization.

Qualitative Visualization. We then compare pre-training and RL post-training in an out-of-distribution interaction case. The pre-trained policy fails to resolve a left-turn conflict between Vehicle 5 and Vehicle 9, resulting in collision at $t = 5.8$ s. After 1M RL steps, the policy learns conservative yielding behavior,

TABLE II
CLOSED-LOOP COMPARISON OF POST-TRAINING METHODS (PRIMARY: CR, OR, AS; SECONDARY: ADE, KIN)

Method	Primary closed-loop objectives			Secondary diagnostics	
	CR (%)↓	OR (%)↓	AS (m/s)↑	ADE (m)↓	Kin (%)↓
Pre-trained only	2.02±0.11	1.68±0.10	8.36±0.42	1.28±0.02	**0.25±0.02**
SFT	1.99±0.07	1.64±0.06	8.37±0.36	**1.15±0.015**	**0.25±0.01**
DPO	1.95±0.13	1.58±0.09	8.15±0.39	1.33±0.04	0.27±0.01
Offline RL	2.16±0.09	1.82±0.14	**8.98±0.68**	1.37±0.05	0.26±0.02
MULTI-ORFT (Online RL)	**1.91±0.12**	**1.36±0.08**	8.61±0.46	1.36±0.04	0.32±0.03

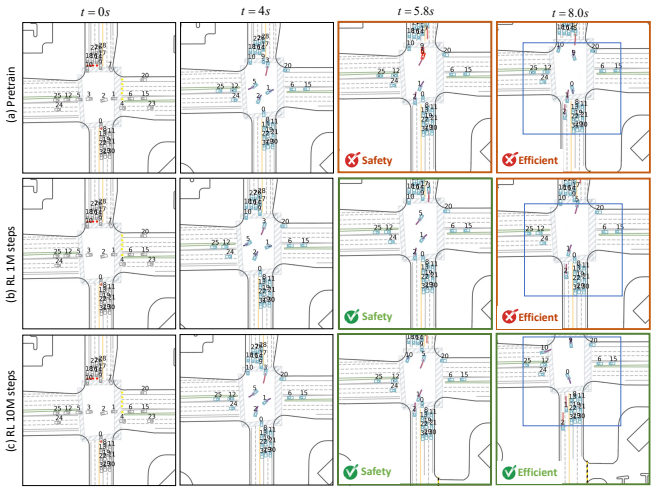


Fig. 4. Qualitative comparison between pre-training and post-training on an 8-second closed-loop rollout from the ego view of Vehicle 1. One-second history trajectories are shown for context.

which improves safety but reduces efficiency. After 10M steps, the policy maintains safety while recovering decisiveness, yielding higher overall traffic progress. This progression shows how reward-driven post-training reshapes interaction strategy.

C. Ablations

Unless otherwise stated, ablations use the same training pipeline and evaluation protocol as Section VI.

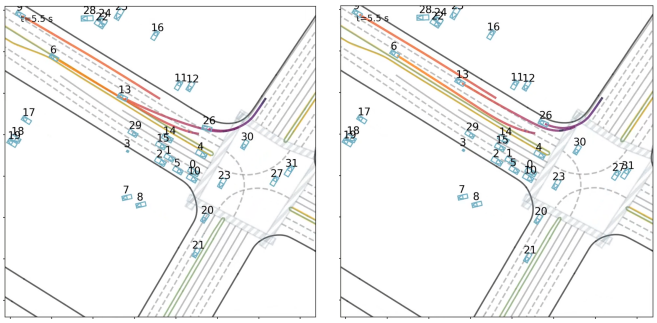
Impact of AdaLN-Zero. AdaLN-Zero improves conditional generation quality. Table III shows gains on multiple metrics, especially off-road rate. The mechanism is direct: condition-driven modulation reshapes denoising features and strengthens road-geometry awareness, which is harder to achieve with cross-attention alone. Better boundary compliance in generation then transfers to improved closed-loop stability.

Fig. 5 provides a qualitative comparison on sharp right-turn cases. Without AdaLN-Zero, Vehicles 6 and 13 drift outward and hit the median boundary. With AdaLN-Zero, turn trajectories are more compact and maintain safer boundary margins.

Impact of Planning/Execution Horizon Ratio. Table IV compares different planning/execution horizon ratios. A longer planning horizon improves lookahead but can weaken short-horizon controllability; in our setting, a moderate ratio gives the best overall balance.

TABLE III
ABLATION ON ADALN-ZERO

Setting	CR (%)↓	OR (%)↓	AS (m/s)↑	ADE (m)↓	Kin (%)↓
w/o AdaLN-Zero	2.11	1.99	**8.40**	1.30	0.26
with AdaLN-Zero	**2.02**	**1.68**	8.36	1.28	0.25



(a) without AdaLN-Zero module (b) with AdaLN-Zero module

Fig. 5. Qualitative comparison with and without the AdaLN-Zero module

TABLE IV
IMPACT OF PLANNING/EXECUTION HORIZON RATIO

T_a/T_p	CR (%)↓	OR (%)↓	AS (m/s)↑	ADE (m)↓	Kin (%)↓
20/20	**0.6587**	0.4190	**0.8014**	**0.8523**	1.7453
20/40	0.6451	0.4166	0.7845	0.8216	1.5712
20/80	0.6448	0.4202	0.7666	0.8387	1.6770

TABLE V
ABLATION ON POST-TRAINING DATA COMPOSITION

Dataset type	CR (%)↓	OR (%)↓	AS (m/s)↑	ADE (m)↓	Kin (%)↓
high-score	1.95	**1.34**	8.53	**1.30**	**0.31**
low-score	2.18	1.99	8.22	1.40	0.37
full	**1.91**	1.36	**8.61**	1.36	0.32

Training Dataset. Table V analyzes post-training data composition. Using only high-score scenes limits trajectory diversity and weakens relative ranking signals. Using only failure scenes overfits local correction and harms global policy balance.

In contrast, the mixed full dataset gives the best overall trade-off, indicating that balanced scenario coverage is critical for stable gradients and robust policy improvement.

VII. CONCLUSION

We introduce MULTI-ORFT for cooperative driving, coupling scene-conditioned diffusion pre-training with online RL fine-tuning. Dual-path conditioning and VG-GRPO in a two-level MDP reduce collision and off-road rates and improve average speed on WOMD. Future work will explore learned objectives and broader evaluation.

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